

ORIGINAL ARTICLE

A machine learning-based analysis of nationwide cancer comprehensive genomic profiling data across cancer types to identify features associated with recommendation of genome-matched therapy

H. Ikushima^{1*}, K. Watanabe^{1,2}, A. Shinozaki-Ushiku^{3,4}, K. Oda³ & H. Kage¹

¹Department of Respiratory Medicine, Graduate School of Medicine, The University of Tokyo, Tokyo; ²Next-Generation Precision Medicine Development Laboratory, Graduate School of Medicine, The University of Tokyo, Tokyo; ³Division of Integrative Genomics, Graduate School of Medicine, The University of Tokyo, Tokyo;

⁴Department of Pathology, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan



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Background: The low probability of identifying druggable mutations through comprehensive genomic profiling (CGP) and its financial and time costs hinder its widespread adoption. To enhance the effectiveness and efficiency of cancer precision medicine, it is critical to identify patient characteristics that are most likely to benefit from CGP.

Patients and methods: This nationwide retrospective study employed machine learning models to predict the identification of genome-matched therapies by CGP, utilizing a national database covering 99.7% of the patients who underwent CGP in Japan from June 2019 to November 2023. Prediction models were constructed for the overall cancer population, specific cancer types, and adolescent and young adult (AYA) group. The SHapley Additive exPlanations (SHAP) algorithm was applied to elucidate clinical features contributing to model predictions.

Results: This study included 60 655 patients [mean age (standard deviation), 60.8 years (14.5 years); 50.1% males]. CGP identified at least one genome-matched therapy in 11 227 cases (18.5%). The best prediction model was eXtreme Gradient Boosting (XGBoost) with an area under the receiver operating characteristic curve of 0.819. Cancer type was the most important predictor (negative for pancreas and positive for breast and lung), followed by the age, presence of liver metastasis, and number of metastatic sites. Analysis of cancer type-specific models identified several organ-specific features, including the sex, interval between the cancer diagnosis and CGP, sampling site, and CGP panel. Among 3455 AYA patients, genome-matched therapies were identified in 459 patients (13.3%). The AYA-specific model achieved an area under the receiver operating characteristic curve of 0.768, with bone tumor identified as a negative predictor in addition to those identified in the overall cancer population model.

Conclusion: Several factors predicting the identification of genome-matched therapies through CGP were identified for the overall cancer population and cancer type-specific subpopulations. Expedited CGP is recommended for patients who match the identified profile to facilitate early targeted therapy.

Key words: comprehensive genomic profiling, genome-matched therapy, machine learning, explainable artificial intelligence, adolescent and young adult

INTRODUCTION

Research and clinical applications of cancer precision medicine, in which therapeutic strategies are customized to fit the unique characteristics of cancer in each patient, have rapidly progressed.¹ Comprehensive genomic profiling (CGP) has played key roles in cancer precision medicine through optimization of therapeutic interventions based on

genomic alterations in cancer cells.^{2,3} This progress has been further boosted by ‘liquid biopsy’, blood-based CGP assays.^{4,5}

However, despite the promise of CGP, several challenges persist. First, the probability of identifying actionable mutations using the CGP is relatively low. Second, the financial cost associated with CGP still presents a significant barrier to its widespread adoption and accessibility. Third, processing of huge amounts of genomic data from CGP is time-consuming and labor-intensive.⁶ Fourth, the sheer volume of genomic data and its increasing complexity make it difficult for health care professionals to understand the full picture without human biases.⁷ These issues underscore the need for complementary strategies to maximize the clinical

*Correspondence to: Dr Hiroaki Ikushima, Department of Respiratory Medicine, Graduate School of Medicine, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8655, Japan. Tel: +81-3-3815-5411
E-mail: hikushima-ty@umin.ac.jp (H. Ikushima).

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utility of CGP for guiding treatment decisions. To address these challenges and optimize the effectiveness and efficiency of cancer precision medicine, it is important to identify the patient characteristics that are most likely to benefit from CGP.

The field of precision oncology has developed in diverse ways. No single institution can treat a sufficient number of patients to improve clinical decision-making independently. To overcome this challenge, some comprehensive pan-cancer clinico-genomic databases, such as The Cancer Genome Atlas (TCGA) and Genomics Evidence Neoplasia Information Exchange (GENIE), have been established.^{8,9} In Japan, the Center for Cancer Genomics and Advanced Therapeutics (C-CAT) database was launched in 2019 and has collected comprehensive clinico-genomic data of >60 000 patients across cancer types, making it one of the largest clinically annotated databases of its kind as a single platform. What makes this database unique is that the registered data are based on routine clinical practice of cancer treatment, not on academia-driven research or clinical trials. Furthermore, almost all patients (99.7%) who underwent CGP in Japan have been covered by the C-CAT database (but only upon patients' consent) according to the regulations of the national health insurance system.^{10,11} Therefore, clinical data registered in the C-CAT database are literally the real-world data.

In the field of handling large amounts of medical data with increasing complexity, the development of artificial intelligence (AI) has ushered in a new era of medical data analysis and interpretation. One of the primary advantages of AI in medical research is its capacity to reveal hidden insights and associations beyond human detection without prespecification. By analyzing multidimensional datasets encompassing patient demographics, genetic profiles, clinical symptoms, and imaging results, AI algorithms can discern complex relationships and enhance personalized treatment strategies.¹² AI works not only as a predictive or analytical tool but also as a generator of new scientific insight via the explainability method.¹³⁻¹⁵

In this study, we explored the clinical features that can predict whether genome-matched therapies are identified using CGP. We first constructed prediction models by training machine learning models using the C-CAT dataset. We then adapted the explainable AI (XAI) approach to the constructed models to identify features of cases in which genome-matched therapies were discovered using cancer CGP. To take into account the characteristic background and feature values of specific cancer categories, similar analyses were conducted for each cancer type and for the adolescent and young adult (AYA) population. To our knowledge, this study represents the first comprehensive analysis of large-scale clinical data obtained from real-world CGP.

PATIENTS AND METHODS

Data sources and study population

Our study used the dataset from the C-CAT, the national data center for cancer CGP in Japan. For each cancer patient

who undergoes CGP, clinical information (Supplementary Table S1, available at <https://doi.org/10.1016/j.esmoop.2024.103998>) and genomic data from the CGP are securely transferred to the C-CAT. Importantly, the Ministry of Health, Labor, and Welfare (MHLW) of Japan indicated that data registration to the C-CAT is a part of medical care under national health insurance (but only upon patients' consent). After receiving the clinical information and genomic data from the CGP tests, the C-CAT generates 'C-CAT Findings' reports containing information of the matched Japanese Pharmaceuticals and Medical Devices Agency (PMDA) or United States Food and Drug Administration (FDA)-approved therapies and clinical trials based on the CGP data. To generate C-CAT Findings reports, a cancer knowledge database for drugs and clinical trials was constructed and is constantly updated within the C-CAT.¹⁰ Each drug is annotated according to the following levels of evidence defined by the C-CAT: A, indicating PMDA and/or FDA-approved drugs for the particular cancer type; B, indicating statistically credible studies and consensus among experts; C, indicating approved drugs or drugs with the positive clinical trials regardless of cancer type; D, indicating that there are case reports suggesting the benefit of drugs regardless of cancer type; E, indicating that the efficacy was reported by *in vivo* and *in vitro* studies; and F, indicating that the genomic alteration was an oncogenic marker. In the present study, the identification of drugs with evidence level A or B through CGP was defined as 'the discovery of genome-matched therapies', which correspond closely to ESMO Scale of Clinical Actionability of molecular Targets (ESCAT) levels tier I (targets ready for implementation in routine clinical decisions) and II (investigational targets that likely define a patient population that benefits from a targeted drug but additional data are needed), although they are not exactly identical.¹⁶

Clinical and genomic data of 60 655 cancer patients were registered in the C-CAT database as of November 2023. Anonymized data were obtained from the C-CAT portal site (https://c-cat_diagnostics.gn.external/start.html). Informed consent was obtained from the cancer patients before they were registered in the C-CAT database.

Study design

This retrospective nationwide study took place over three major sections: the analysis of the overall cancer population, those with specific cancer types, and the AYA age groups. In each section, we first constructed and trained a machine learning model that used clinical information available before CGP as input (Supplementary Table S1, available at <https://doi.org/10.1016/j.esmoop.2024.103998>) and predicted whether or not one or more approved drugs would be discovered through CGP. The input data included all clinical information registered in the C-CAT, except histopathology and administered chemotherapy. Next, we extracted significant features that contribute to the model's predictions to identify features that predict cases in which genome-matched therapies are

discovered using CGP. This study was approved by the Research Ethics Committee of the Faculty of Medicine, the University of Tokyo (2021341G) and the Information Utilization Review Board of C-CAT (CDU2022-026N).

Model development and statistical analyses

We used Python (version 3.9.5) to develop our machine-learning models. The dataset was split into 80% training set and 20% test set. We constructed and trained random forest classifier, extreme gradient boosting (XGBoost), and category boosting (CatBoost) machine learning models to create prediction models for identification of approved drugs through CGP.¹⁷⁻¹⁹ For cases in which prior cancer type-specific companion diagnostic tests were not carried out, missing values were filled in with 'Unknown'. Intervals between the diagnosis and CGP or between specimen collection and CGP were calculated using the registration, diagnosis, and specimen collection dates. If required for input into the model, categorical features were encoded using a one-hot encoding scheme, which created binary columns for each category. In the training by the five-fold cross validation method, the training dataset was stratified by the class label and divided into five folds. The hyperparameters were optimized using Optuna (version 3.4.0).²⁰ To evaluate the trained models, we assessed the area under the receiver operating characteristic curve (AUROC) for each classifier using the holdout test dataset. The difference in the proportion of drug identification was tested using the chi-square test. All source codes for model development are available at <https://github.com/hiroikushi/CCAT>.

Explainability analyses

To identify features that contribute to model prediction, we applied the SHapley Additive exPlanations (SHAP) algorithm (version 0.43.0) to the constructed machine learning models.²¹ This approach allocates a numerical value (SHAP value) to each feature to measure its impact on the model output and provide a local explanation. A positive SHAP value increases the likelihood of discovery of genome-matched therapies, whereas a negative SHAP value reduces the likelihood of discovery. The SHAP values for each feature and each case in the test dataset were calculated, and the features were arranged in descending order of the mean of absolute SHAP values.

RESULTS

Overview of the cohort

Clinical information of 60 655 patients with solid tumors was registered in the C-CAT database between June 2019 and November 2023. Among them, the 10 most common cancer types (OncoTree Level 1) were bowel (10 182), pancreas (8691), biliary tract (5062), breast (3777), esophagus/stomach (3756), prostate (3717), ovary/fallopian tube (3541), lung (3470), soft tissue (2580), and uterus (2227) (Figure 1A, Table 1). As of November 2023, three tissue-

based and two blood-based CGP tests have been approved in Japan: FoundationOne CDx (44 927, 74.1%), OncoGuide National Cancer Center (Japan) OncoPanel (6807, 11.2%), GenMineTOP (29, <0.1%), FoundationOne Liquid CDx (8819, 14.5%), and Guardant360 CDx (73, 0.1%).²² Among all registered patients, at least one genome-matched therapy was identified in the C-CAT Findings reports for 11 227 patients (18.5%) based on CGP results (Figure 1B).

Construction of prediction models for discovery of genome-matched therapies

To construct machine learning models that can predict whether or not at least one genome-matched therapy is likely to be discovered through CGP, we trained three types of models: random forest classifier, XGBoost, and CatBoost. The clinical information available before CGP (Supplementary Table S1, available at <https://doi.org/10.1016/j.esmoop.2024.103998>) was used as the input for these models. The three trained models achieved similar AUROCs ranging from 0.802 to 0.819 in the holdout test dataset (Figure 1C). Of these three models, we focused on the XGBoost model, which had the highest AUROC (sensitivity 0.799, specificity 0.692), for subsequent analyses. The model demonstrated good calibration; the mean predicted probability was similar to the observed discovery rate of genome-matched therapies in the test dataset (Figure 1D). In addition, similar performance was observed when an XGBoost model was trained using the five-fold cross validation method (Supplementary Figure S1, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). Although we also constructed an XGBoost model using only tissue CGP data to exclude the impact of false negatives in liquid CGP, the tissue-only model showed almost similar performance to that of the model developed using the entire cohort (data not shown).

Next, we evaluated the features that contributed to model prediction using SHAP plots (Figure 1E). Among the clinical information available before undergoing CGP, cancer type was the most important feature for predicting whether genome-matched therapies would be identified. In particular, positive SHAP values were observed in patients with breast, lung, prostate, and bowel cancers, whereas negative SHAP values were associated with pancreatic, brain, and biliary tract cancers. Advanced age and distant metastases (liver metastasis and number of metastatic sites) were also associated with positive SHAP values.

Construction of cancer type-specific prediction models

The possibility of discovering genome-matched therapies varied considerably across different cancer types (Figure 2A). The finding that cancer type is the most important feature prompted us to construct cancer type-specific prediction models and identify crucial cancer type-specific clinical features. Therefore, we trained the random forest classifier, XGBoost, and CatBoost using a cancer type-specific dataset. In addition to the clinical

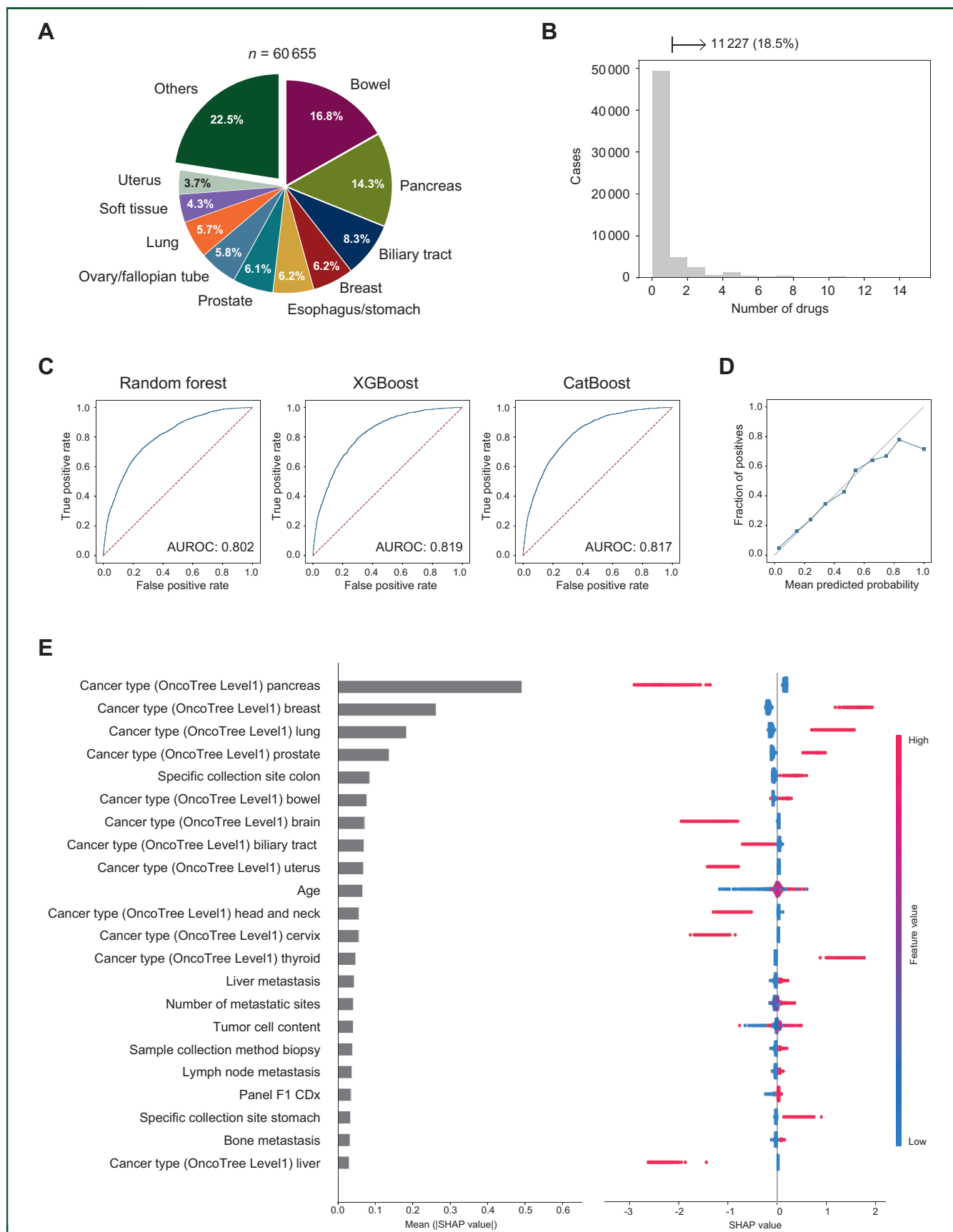


Figure 1. Prediction models for the whole C-CAT cohort. (A) Distribution of cancer types among 60 655 patients registered in the C-CAT database. The 10 most frequent cancer types are shown in the pie chart. (B) Distribution of the number of identified drugs with evidence level A or B in the C-CAT Findings Report. (C) The receiver operating characteristic curve of the constructed models for the holdout test set. (D) The calibration plot visualizes how correctly the probability predicted by the XGBoost model estimates the observed discovery rate of genome-matched therapies in the test dataset. The x-axis of each point represents the mean predicted

features utilized in the pan-cancer prediction models, cancer type-specific clinical information registered in the C-CAT database was included as input data for the training of cancer type-specific prediction models: epidermal growth factor receptor (*EGFR*) mutation, anaplastic lymphoma kinase (*ALK*) translocation, c-ros oncogene 1 (*ROS1*) translocation, B-Raf proto-oncogene (*BRAF*) V600 mutation, programmed death-ligand 1 (PD-L1) expression, and asbestos exposure history for lung cancer; estrogen receptor (ER) expression, progesterone receptor (PgR) expression, human epidermal growth factor receptor type 2 (HER2) expression, and germline breast cancer susceptibility gene (*BRCA*) 1/2 mutation for breast cancer; Kirsten rat sarcoma viral oncogene homolog (*KRAS*) mutation, neuroblastoma RAS homolog (*NRAS*) mutation, *BRAF* V600 mutation, HER2 expression for esophageal, gastric, and colorectal cancers (Supplementary Table S1, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). For each cancer type, three kinds of machine learning models showed comparable predictive power in the holdout test dataset (Figure 2B). Therefore, we focused on the XGBoost model for subsequent analyses, like the pan-cancer prediction model.

The AUROC of the prostate cancer-specific models was almost 0.5, indicating no discriminating ability. The proportion of cases with a positive label (i.e. patients for whom at least one drug with evidence level A or B was identified through CGP) was <5% in both the pancreas and uterus datasets. Therefore, subsequent studies were carried out using prediction models specific to each of the seven cancer types other than the prostate, pancreas, and uterus.

Cancer type-specific features in prediction models for discovery of genome-matched therapies

To identify cancer type-specific clinical features that increase the possibility of identifying genome-matched therapies, we next analyzed features that contribute to the prediction of cancer type-specific models.

Bowel. Combination therapy with *BRAF* inhibitors, mitogen-activated protein kinase kinase (MEK) inhibitors, and cetuximab is categorized as evidence level A for colorectal cancer patients with a *BRAF* V600E mutation,²³ and a combination of trastuzumab and pertuzumab for patients with HER2 overexpression.²⁴ Combination therapy of irinotecan and selumetinib is categorized as evidence level B for colorectal cancer patients with a *KRAS* exon 2 (codon 12 and 13) mutation,²⁵ but no drug is categorized as evidence level A or B for patients with *KRAS* exon 3 (codons 59 and 61) or exon 4 (codons 117 and 146) mutations. Furthermore, sotorasib and adagrasib were approved in combination with anti-EGFR antibodies for colorectal cancer patients with *KRAS* G12C mutation by FDA.^{26,27} Therefore, it is

reasonable that positive SHAP values were associated with a *BRAF* mutation, HER2 expression 3+, and *KRAS* codon 12 and 13 mutations, whereas negative SHAP values were associated with a *KRAS* codon 146 mutation (Figure 3A). In addition, positive SHAP values were associated with the presence of lung, lymph node, or liver metastasis and advanced age in the bowel cancer-specific prediction model, whereas negative SHAP values were associated with the interval between the cancer diagnosis and CGP.

Biliary tract. The information that the sample was collected from the liver was linked to positive SHAP values in the biliary tract-specific prediction model (Supplementary Figure S2A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). This result can be clinically interpreted in two ways: first, cases with liver metastasis may have a higher probability of genome-matched therapies being identified through CGP. In fact, liver metastasis was also associated with positive SHAP values in this prediction model (Supplementary Figure S2A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). The other interpretation is that patients with intrahepatic cholangiocarcinoma may have a higher probability of genome-matched therapies being identified, which can be confirmed by a direct comparison between intrahepatic cholangiocarcinoma and other biliary tract cancers (Supplementary Figure S2B, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). In addition, positive SHAP values were associated with metastasis to the bone, lymph node, or lung, sample collection by biopsy, tumor cell content, number of metastatic sites, and a family history of cancer, while negative SHAP values were associated with interval between the diagnosis and CGP and male (Supplementary Figure S2A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>).

Breast. Positive SHAP values were associated with advanced age and hormone receptor- and/or HER2-positive in immunohistochemistry, whereas negative SHAP values were associated with intervals between specimen collection and CGP (Figure 3B). Positive SHAP values were also associated with the number of metastatic sites and presence of bone, liver, or brain metastases. The frequency of the detection of circulating tumor DNA (ctDNA) was reported to be higher in breast cancer patients with distant metastasis.²⁸ Therefore, the higher ctDNA detection rate in patients with distant metastasis than in those without it among those who underwent CGP on blood samples may be one reason why distant metastasis is associated with a positive SHAP value. Another possible explanation is that distant metastasis is an important predictor, independent of the ctDNA detection rate. To test this hypothesis, we constructed another breast-specific XGBoost prediction model after excluding cases in which 'Specimen type' is 'Peripheral blood' in the database.

probability in each bin, whereas the y-axis of each point represents discovery rate of genome-matched therapies in each bin. The dashed line represents a perfect calibrated model. (E) SHAP summary plot of the most contributing features ranked by descending contribution to the prediction of the whole cohort model. The left bar chart represents the mean absolute SHAP values across the cases in the holdout test set. In the right chart, the colors represent the feature values, with red indicating higher or positive values and blue indicating lower or negative values.

C-CAT, Center for Cancer Genomics and Advanced Therapeutics; SHAP, SHapley Additive exPlanations.

Table 1. Patient characteristics in the C-CAT cohort

Cancer type (OncoTree level 1)	n	Sex			Age, years			Panel				
		Female	Male	Not entered/ unknown	Mean	SD	Median	F1 CDx	NCC	GenMineTOP	F1 liquid	Guardant360
Total	60 655	30 250	30 400	5	60.8	14.5	63.0	44 927	6807	29	8819	73
Bowel	10 182	4656	5526	0	60.7	12.0	62.0	8207	1040	4	921	10
Pancreas	8691	3819	4871	1	64.8	10.3	66.0	4884	1558	3	2226	20
Biliary	5062	1968	3094	0	67.4	10.0	69.0	3429	623	3	998	9
Breast	3777	3750	27	0	55.6	11.4	55.0	2746	414	0	611	6
Esophagus/stomach	3756	1026	2730	0	63.2	11.7	65.0	2857	492	1	402	4
Prostate	3717	2	3717	0	71.7	7.8	73.0	2112	207	0	1391	7
Ovary/fallopian tube	3541	3541	0	0	57.4	11.6	57.0	3018	296	3	224	0
Lung	3470	1286	2184	0	63.6	11.7	66.0	2253	376	1	831	9
Soft tissue	2580	1204	1374	2	53.0	19.9	57.0	2202	243	5	130	0
Uterus	2227	2226	1	0	60.1	10.2	60.0	1962	184	1	80	0

Ten most frequent cancer types were shown in the table.

C-CAT, Center for Cancer Genomics and Advanced Therapeutics; NCC, National Cancer Center (Japan) OncoPanel.

In this tissue biopsy-only prediction model, the number of metastatic sites and the presence of bone or liver metastasis were still associated with positive SHAP values (Supplementary Figure S3, available at <https://doi.org/10.1016/j.esmoop.2024.103998>), indicating that distant metastases, such as bone and liver, are important independent predictors for the identification of genome-matched therapies.

Esophagus/stomach. Positive SHAP values were associated with HER2 positivity in immunohistochemistry, advanced age, and metastasis to the liver, lymph nodes, or lungs (Supplementary Figure S4A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). Similar to the biliary tract- or breast-specific prediction models (Figure 3B, Supplementary Figure S2A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>), the information that the sample was analyzed using FoundationOne CDx was linked to positive SHAP values in the esophagus/stomach-specific prediction model. Indeed, the frequency of identification of genome-matched therapies was higher in cases analyzed by FoundationOne CDx than in those analyzed by another tissue-based CGP (OncoGuide National Cancer Center [Japan] OncoPanel) or those analyzed by a blood-based CGP (FoundationOne liquid CDx) (Supplementary Figure S4B, available at <https://doi.org/10.1016/j.esmoop.2024.103998>).

Ovary/fallopian tube. Positive SHAP values were associated with a family history of cancer, specimen collection from metastatic lesions, metastasis to the peritoneum or liver, and the presence of double cancer in the ovary/fallopian tube-specific model (Supplementary Figure S5, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). Hereditary breast and ovarian cancer (HBOC) can be related to a family history and the presence of double cancer.

Lung. EGFR tyrosine kinase inhibitors, such as osimertinib, are categorized as evidence level A in lung cancer patients with EGFR exon 19 or 21 alterations²⁹ and amivantamab for patients with EGFR exon 20 insertion.³⁰ It has been

reported that these EGFR mutations were more prevalent in female lung cancer patients with a history of minimal or no smoking.³¹ Therefore, it is reasonable that positive SHAP values were found to be associated with EGFR mutations, whereas negative SHAP values were associated with smoking history and male (Figure 3C). In addition, analysis using FoundationOne CDx and metastasis to the bone, brain, or lung were also associated with positive SHAP values.

Soft tissue. The specimen collection site, which may reflect a primary or metastatic tumor site, is a crucial predictor in the soft tissue-specific model (Supplementary Figure S6A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). Specifically, collection from the peritoneum, stomach, liver, or small intestine was associated with positive SHAP values, whereas collection from the skin or lungs was associated with negative SHAP values. Imatinib, sunitinib, and regorafenib are categorized as evidence level A for gastrointestinal stromal tumor (GIST) with particular KIT or PDGFRA mutations.³² Genome-matched therapies were identified more frequently when samples were obtained from the stomach, small intestine, peritoneum, or liver (Supplementary Figure S6B, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). The peritoneum and liver were also associated with positive SHAP values as metastasis sites (Supplementary Figure S6A, available at <https://doi.org/10.1016/j.esmoop.2024.103998>).

Construction and analyses of the AYA-specific prediction model

In the whole cancer population prediction model and some cancer type-specific models, advanced age is one of the positive predictive indicators for the discovery of genome-matched therapies, which means that CGP is less likely to identify drugs in younger patients than in older ones. Therefore, to maximize the clinical utility of CGP in guiding cancer treatment decisions in younger patients, we constructed an AYA-specific prediction model. In the C-CAT

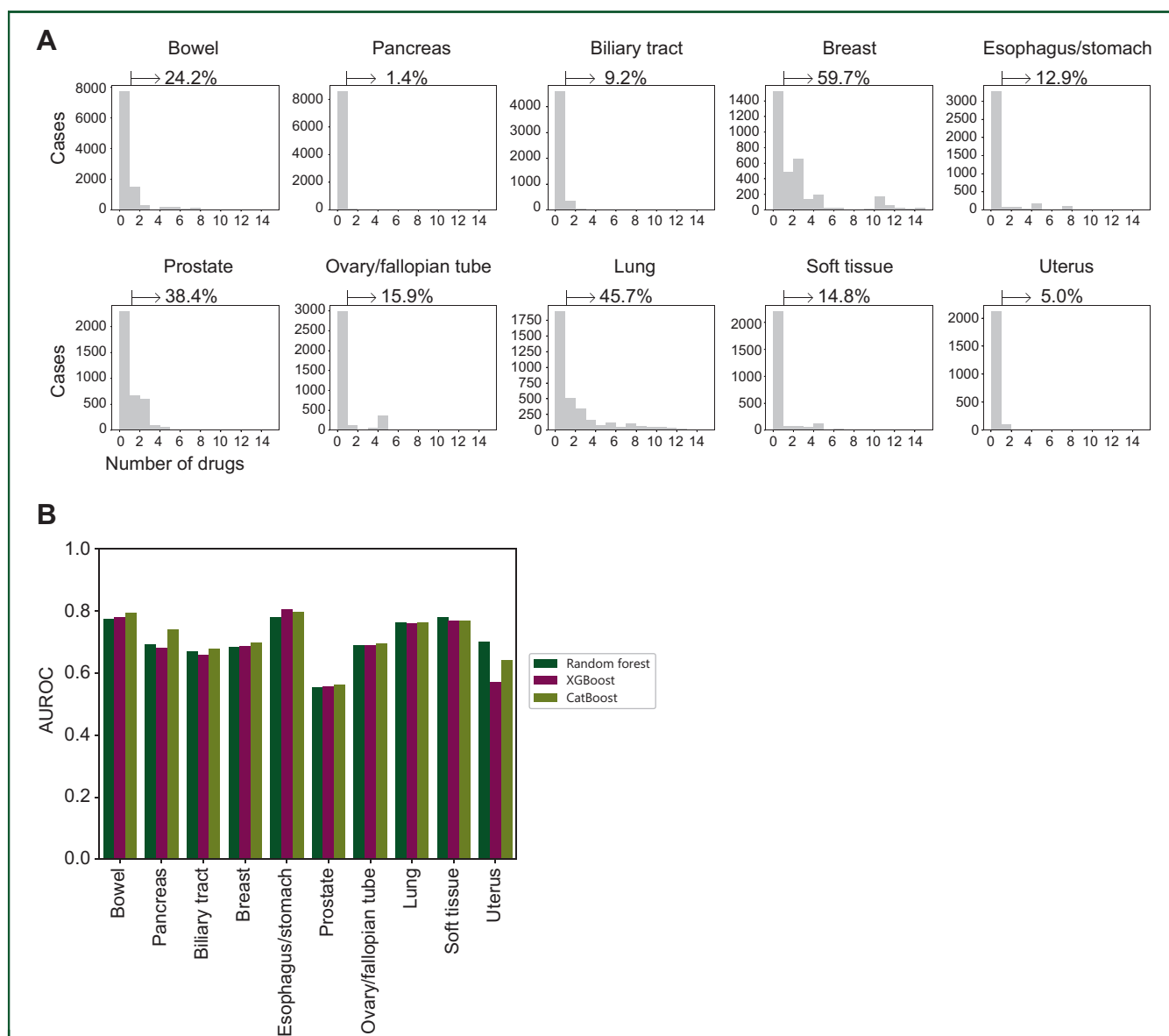


Figure 2. Cancer type-specific trends in the identification of genome-matched therapies. (A) Distribution of the number of identified drugs with evidence level A or B in each cancer type-specific cohort. The proportion of cases in which at least one drug with evidence level A or B was identified is shown on each graph. (B) AUROC of the trained cancer type-specific random forest-, XGBoost-, and CatBoost-based prediction models for the holdout test dataset. AUROC, area under the receiver operating characteristic curve.

database, 3455 AYA patients (15-39 years old) were registered, and among them, at least one genome-matched therapy was discovered by CGP in 459 patients (13.3%, Figure 4A). We trained an XGBoost machine learning model, which achieved an AUROC of 0.768 (sensitivity 0.769, specificity 0.662) in the holdout test dataset after training (Figure 4B). Similar performance was observed when an XGBoost model was trained using the five-fold cross validation method (Supplementary Figure S7, available at <https://doi.org/10.1016/j.esmoop.2024.103998>). The SHAP plot for the AYA-specific prediction model showed that the crucial factors for this group were generally similar to those for the overall population (Figure 4C): cancer type, age, and distant metastasis such as to the liver. Positive SHAP values were observed in patients with breast, bowel, and lung

tumors, whereas negative SHAP values were observed in patients with cervix, soft tissue, and bone tumors.

DISCUSSION

In this study, using a machine learning-based analysis of a nationwide large-scale clinical CGP database, we elucidated predictors for the identification of genome-matched therapies through CGP. Using clinical information obtained before CGP as input data, we developed machine learning models to predict whether genome-matched therapies could be identified through CGP in a specific patient. The XGBoost-based model showed good calibration between the predicted probability and the observed discovery rate of genome-matched therapies. The XAI approach to the

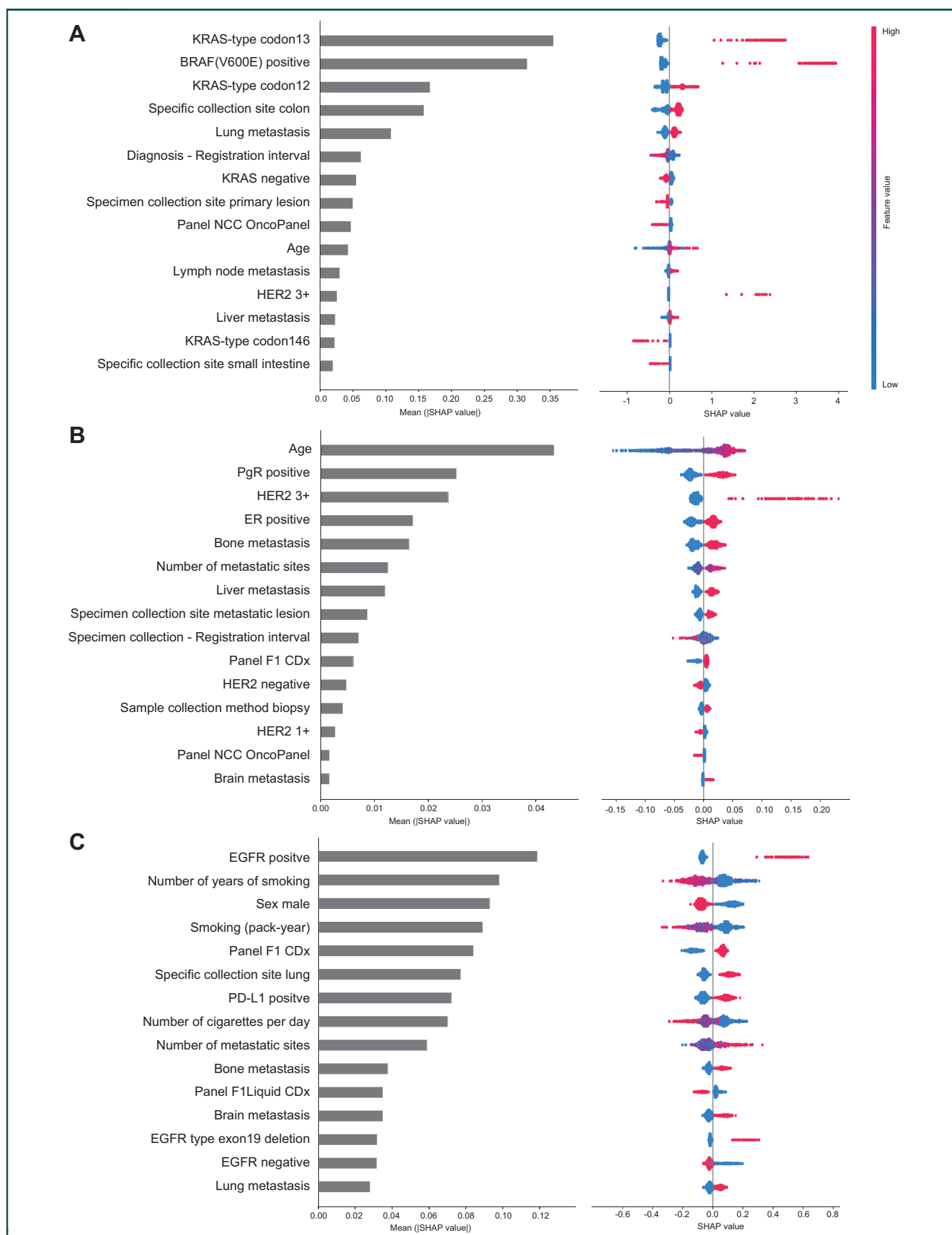


Figure 3. Explainability analysis of the cancer type-specific prediction models. SHAP summary plot of the most contributing features in the (A) bowel-, (B) breast-, and (C) lung-specific prediction models.

BRAF, B-Raf proto-oncogene; EGFR, epidermal growth factor receptor; ER, estrogen receptor; HER2, human epidermal growth factor receptor type 2; KRAS, Kirsten rat sarcoma viral oncogene homolog; NCC, National Cancer Center (Japan) OncoPanel; PD-L1, programmed death-ligand 1; PgR, progesterone receptor; SHAP, SHapley Additive exPlanations.

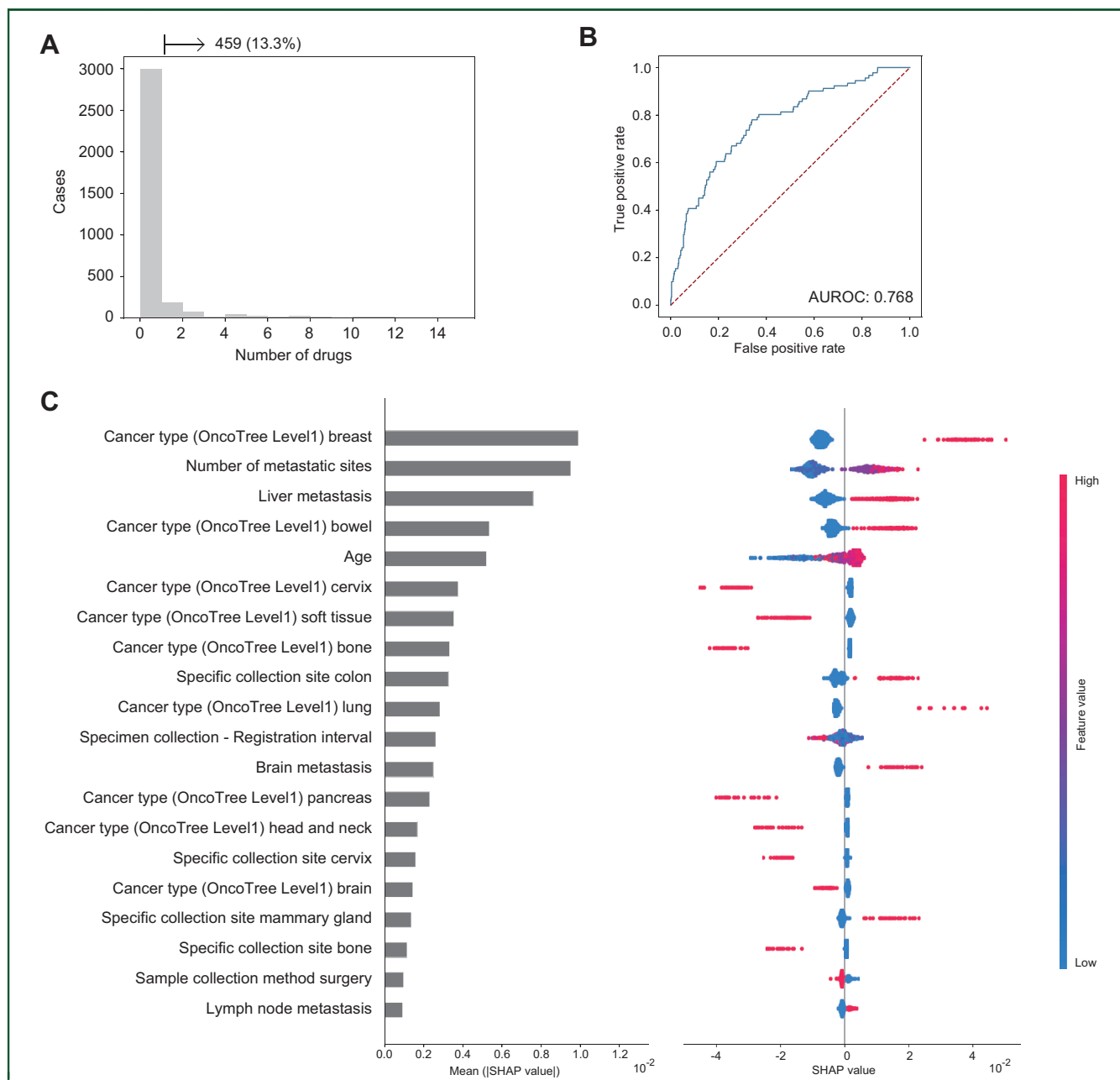


Figure 4. Construction and analyses of the AYA-specific prediction model. (A) Distribution of the number of identified drugs with evidence level A or B in the AYA cohort. (B) The receiver operating characteristic curve of the constructed model for the holdout test set. (C) SHAP summary plot of the most contributing features in the AYA-specific prediction model.

AUROC, area under the receiver operating characteristic curve; AYA, adolescent and young adult.

constructed model revealed the importance of cancer type, age, and distant metastasis in the discovery of genome-matched therapies. Based on this analysis, we also constructed cancer type-specific and AYA-specific prediction models and identified crucial predictors for these subgroups.

Previous studies have analyzed patient data to identify important predictors for the discovery of genome-matched therapies using CGP.³³⁻³⁶ These studies, however, analyzed CGP data in small cohorts or specific cancer types to evaluate the contribution of predetermined clinical factors. In our study, almost all registered clinical information available before the CGP was used as input data for machine learning

models without any prespecification. This analytical strategy allowed us to conduct a comprehensive analysis of clinical data, thereby enabling us to obtain results with objective ranks.

In the context of perioperative precision therapy, some studies have reported the usefulness of a liquid biopsy in consideration of adjuvant chemotherapy.^{37,38} In these studies, the quantity of ctDNA after surgery may reflect the hematogenous spread of the tumor at the time of surgery. In our study, the number of metastatic organs and the presence of specific metastases were identified as positive predictors in the pan-cancer model and some cancer type-specific models. We reconstructed a tissue-only prediction

model for breast cancer, for which the presence of distant metastasis was a crucial positive predictive indicator, and showed that distant metastasis was also a positive predictor in the tissue-only model. These results indicate that the presence of distant metastasis reflects not only the quantity of ctDNA, but also the accumulation of driver mutations in tumor cells.

In some cancer type-specific models, the interval between specimen collection and C-CAT registration (i.e. the timing of CGP) or between the diagnosis and C-CAT registration was identified as a negative predictor. There are several possible clinical reasons for these results. First, long-term storage of specimens results in DNA/RNA degradation, leading to a lower detection rate of tumor genomic alterations.³⁹⁻⁴¹ Second, tumor aggressiveness may reflect the accumulation of genomic alterations in the tumors. It should be noted that CGP is covered by the Japanese national health insurance only for advanced solid tumors in patients who have completed or will soon complete standard therapies. Therefore, a short interval between the diagnosis and CGP represents not only rapid tumor progression, but also strong resistance to standard chemotherapy. In addition to our analysis, a previous study on CGP for non-small-cell lung cancer patients in the USA demonstrated that timely CGP before first-line initiation was associated with increased targeted therapy use, decreased expenditure on immune checkpoint inhibitors, and a longer time to therapy discontinuation.⁴² Therefore, timely CGP is an important diagnostic strategy for delivering genome-matched therapies to adequate patients irrespective of the health insurance system.

A few studies have reported the statistics of CGP tests for AYA patients. Morfouace et al.⁴³ analyzed the results of CGP tests for patients 12-29 years old with intermediate- or high-grade bone and soft tissue sarcomas. Of the 48 patients in their study, CGP testing led to therapeutic recommendations in 81%. Andrew et al.⁴⁴ reported that potentially actionable variants were detected in 14 (56%) of 25 patients 15-39 years old with sarcomas. In another study, OncoKids, a comprehensive DNA- and RNA-based next-generation sequencing panel, was reported to identify targetable genomic alterations in 18% of patients 0-39 years old with central nervous system tumors.⁴⁵ Age ranges and definitions of drug identification vary among studies, however, making it difficult to compare results across these studies. Advanced age was identified as one of the positive predictors for the discovery of genome-matched therapies in our analysis of the overall cancer population, and the frequency of identification of genome-matched therapies was lower in AYA patients. To our knowledge, our study is the first to analyze the results of CGP in a large pan-cancer cohort of AYA patients. As in the analysis of the whole population, cancer type is a crucial predictive factor for the identification of genome-matched therapies in AYA patients.

Recent research developments in deep learning have enabled AI systems to predict the presence of driver

mutations using histopathology or computed tomography.^{46,47} The combination of such systems and our prediction strategy may synergistically reinforce the efficiency of CGP. Importantly, the role of CGP is not limited to the identification of matched therapies, but also includes the refinement of diagnosis. Since our study as well as C-CAT findings reports do not cover other contributions of CGP than the identification of genome-matched therapies, future studies are required to construct prediction models for refining diagnosis and selecting standard therapies other than genome-matched therapies.

Several limitations associated with the present study warrant mention. There was a lack of prospective validation of our prediction model. Almost all of the patients registered in the C-CAT database were Asian, which prevented the analysis of patient ethnicity. Since this study analyzed the C-CAT database retrospectively, some clinically important features, such as companion diagnostic results, have missing values. In particular, histopathological information has missing values in many cases with particular cancer types; thus, it was not included as input data in the prediction models. The distribution of patients registered in the C-CAT database differs from the total population diagnosed with cancer, as patients in whom druggable driver mutations have been identified by cancer type-specific small companion diagnostic tests may not undergo CGP. Patients who were unable to receive additional treatments after standard chemotherapy owing to a poor performance status or any other reasons were also excluded. In the model training, the distribution of class labels was unbalanced. To deal with unbalanced data, we assigned a higher penalty to misclassifications of the minority class using the *scale_pos_weight* parameter in XGBoost.

In conclusion, we developed machine learning models that use clinical information as input data and predict the likelihood of identifying genome-matched therapies through CGP. An XAI analysis of the constructed models identified several predictors for the overall cancer population as well as cancer type-specific subpopulations. Expedited CGP is recommended for patients who match the identified profile to facilitate early targeted therapy.

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